Meghana Reddy L

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Data Engineer

PROFESSIONAL SUMMARY:

- Data Engineer with 6 years of experience designing scalable ETL/ELT pipelines using PySpark, SQL, and Python across AWS, Azure, and GCP for enterprise-grade data processing and analytics.
- Strong expertise in GCP services including BigQuery, Dataflow, Pub/Sub, Cloud Storage, and Composer to build highly available, real-time and batch pipelines for large-scale structured and semi-structured data.
- Reduced real-time processing latency by 30% using Pub/Sub and Cloud Composer with Apache Beam, enabling automated event-driven workflows and streaming analytics across time-sensitive enterprise datasets.
- Successfully migrated legacy systems to cloud platforms (AWS, Azure, GCP), improving performance by 45% and cutting infrastructure costs by 20% through managed services and cloud-native optimizations.
- Optimized Spark-based data transformations using Delta Lake on Databricks to process 1B+ records daily, reducing batch runtime from 12 hours to under 1 hour with adaptive execution tuning.
- Built streaming ingestion frameworks using Apache Kafka, AWS Kinesis, and Azure EventHub integrated with Databricks and Spark Streaming for event-driven architectures in banking and healthcare domains.
- Delivered secure data lakes using AWS S3 and Azure Data Lake Gen2 with fine-grained access via IAM, RBAC, and Key Vault, ensuring encryption at rest and in transit.
- Engineered parameterized, reusable pipelines in Azure Data Factory to automate ingestion from SQL Server, Oracle, and SFTP across 20+ datasets, enabling dynamic orchestration and centralized metadata control.
- Worked with structured and semi-structured formats including JSON, Avro, Parquet, and CSV from APIs, NoSQL systems like Cosmos DB, and file systems, integrating into scalable cloud data platforms.
- Designed and maintained dimensional models (Star, Snowflake) for fact and dimension tables in BigQuery and Azure Synapse, supporting performant BI reporting and historical data analysis.
- Implemented CI/CD pipelines using Azure DevOps, GitHub Actions, and Terraform, standardizing infrastructure as code and improving data platform deployment velocity by over 40% across multiple environments.
- Automated data quality validation, schema enforcement, and lineage using Great Expectations, GCP Data Catalog, and Stackdriver, increasing governance and data reliability across production pipelines.
- Delivered executive dashboards using Spark Streaming and Databricks, integrating with business KPIs and reducing decision latency by 50% through real-time insights and operational metrics.
- Integrated GCP BigQuery datasets with Power BI and Tableau to deliver responsive, self-service dashboards and perform complex ad hoc analysis across finance, compliance, and product teams.
- Collaborated with product owners, analysts, and engineers to define scalable data architectures that translate business goals into secure, cloud-native data platforms with compliance and analytics built-in by design.

TECHNICAL SKILLS:

- **Cloud Platforms -** AWS (S3, Glue, Kinesis, Redshift, EMR, Step Functions, Lambda, DMS, Lake Formation, KMS, IAM, CodePipeline, CloudFormation, MWAA), Azure (Data Factory, Databricks, Synapse Analytics, Event Hub, Key Vault, ADLS Gen2, Analysis Services, Azure Monitor, Log Analytics, Azure DevOps), GCP (BigQuery, Dataflow, Pub/Sub, Cloud Storage, Composer, Data Catalog)
- **Big Data/Processing -** PySpark, Apache Spark, Delta Lake, Apache Beam, Apache Kafka, Spark Structured Streaming, Databricks, EMR, ETL/ELT
- **Languages & Tools -** Python, SQL, Pytest, Pandas, Terraform, Git, Azure DevOps, GitHub Actions, Jenkins, Great Expectations
- **Databases & Warehousing -** BigQuery, Azure Synapse Analytics, AWS Redshift, Oracle, PostgreSQL, SQL Server, Cosmos DB (NoSQL)
- **Data Modeling & BI -** Dimensional Modeling (Star Schema, Snowflake Schema), SCD Type 2, Power BI, Tableau, Amazon QuickSight, Azure Analysis Services
- **Formats & Data Governance -** JSON, Avro, Parquet, CSV, Metadata Registries, Lineage Tracking, Schema Enforcement, CI/CD

PROFESSIONAL EXPERIENCE:

Pfizer **July 2024 – Present**

Data Engineer

Responsibilities:

- Designed secure, scalable AWS data lakes using S3, Glue, and Lake Formation to ingest, catalog, and transform clinical trial data while supporting enterprise-scale compliance and access control requirements.
- Built PySpark-based ETL pipelines on AWS Glue that automated ingestion of over 30 structured and semi-structured sources, enabling consistent and reusable processing across global drug discovery pipelines.
- Implemented near real-time streaming pipelines using AWS Kinesis and Spark Structured Streaming to support event-driven processing of clinical trial events with high reliability and low latency.
- Engineered large-scale genomic transformation pipelines on AWS EMR using PySpark to process over 2 TB of daily data, optimizing shuffle partitions and execution time across clusters.
- Orchestrated end-to-end workflows with AWS Step Functions and Lambda functions to coordinate batch and streaming tasks across Glue, Redshift, and S3 storage layers.
- Created optimized Redshift clusters with WLM queues, tuned distribution keys, and implemented compression to improve query execution time by 40% and increase data warehouse efficiency.
- Developed change data capture (CDC) pipelines with AWS DMS and Glue for real-time replication of updates from clinical source systems into Redshift for near real-time analytics.
- Managed schema evolution and metadata through Glue Schema Registry and AWS MSK, while maintaining discoverability using Glue Data Catalog for consistent schema enforcement and lineage.
- Implemented data security protocols using AWS KMS and IAM role-based access, applying encryption at rest and in transit across S3 buckets and Redshift warehouses for compliance.
- Wrote Pytest-based unit tests for PySpark transformations covering 80%+ of logic paths, ensuring data integrity and test-driven reliability in production-grade data pipelines.
- Deployed infrastructure and pipelines using AWS CodePipeline and CloudFormation templates, eliminating manual provisioning and reducing deployment errors by over 90%.
- Developed interactive executive dashboards using Athena and Amazon QuickSight to visualize clinical metrics and KPIs in near real-time for leadership decision-making.
- Scheduled, orchestrated, and monitored workflows using Apache Airflow (MWAA) to automate daily pipeline executions, enforce SLAs, and enable recovery from job failures.
- Partnered with pharmacovigilance teams to ensure AWS-hosted pipelines met regulatory standards, and built SageMaker preprocessing pipelines to accelerate ML-driven disease progression prediction models.

Bank of America **January 2022 – June 2024**

Data Engineer

Responsibilities:

- Designed modular Azure Data Factory pipelines with parameterized datasets and metadata-driven orchestration to integrate financial data from 50+ systems, enabling high-throughput data ingestion and transformation.
- Built Spark-based transformation pipelines in Azure Databricks to process over 500 million transactions daily, applying custom PySpark UDFs to normalize and standardize complex financial and transactional data.
- Engineered Delta Lake solutions to support ACID compliance and time-travel features for historical and regulatory reporting on customer and transactional datasets within Databricks.
- Integrated Azure Event Hub with Databricks Structured Streaming and Kafka feeds to power real-time fraud detection use cases and financial transaction analytics across high-volume data streams.
- Designed and deployed dimensional models in Azure Synapse Analytics using Star/Snowflake schemas to support enterprise BI dashboards for audit, compliance, and risk analysis teams.
- Applied SCD Type 2 logic within PySpark pipelines and maintained slowly changing dimension tables to track versioned transaction histories across critical customer data sources.
- Used adaptive query execution (AQE), dynamic partition pruning, and Spark optimization techniques to improve processing performance by 35% across streaming and batch workloads.

- Implemented robust security practices using Azure Key Vault, RBAC, and service principals to manage credentials, access controls, and compliance with internal security guidelines.
- Monitored production pipelines with Azure Monitor and Log Analytics, building SLA-based alerts and telemetry-driven diagnostics to improve workflow reliability and early error detection.
- Automated CI/CD deployments with Azure DevOps, conducting peer code reviews, enforcing quality gates, and standardizing rollback-ready data pipeline deployment procedures.
- Built Power BI dashboards integrated with Synapse and Azure Analysis Services to provide daily insights into financial risk metrics and transaction anomalies for leadership teams.
- Mentored junior data engineers on Spark optimization, dimensional modeling, and Azure best practices, driving team adoption of cloud-native tooling and enterprise-scale data standards.

Programobil **July 2019 – July 2021**

Data Engineer

Responsibilities:

- Designed scalable ETL workflows in Python and SQL to ingest product lifecycle, ERP, and CRM data into Oracle and PostgreSQL systems using Airflow- and cron-based scheduling.
- Built star-schema models with fact and dimension tables, supporting Tableau reporting for business users and improving data visibility for inventory and manufacturing performance metrics.
- Developed Spark and PySpark pipelines to process telemetry data from factory devices for predictive maintenance, leveraging Kafka producers and real-time ingestion streams.
- Transformed data from CSV, JSON, and Parquet formats using pandas and PySpark, preparing unified datasets from flat files and APIs for analytics.
- Tuned complex SQL queries with indexing and partitioning strategies, reducing reporting refresh times from 8 minutes to under 1 minute across operational dashboards.
- Built robust data validation scripts and reconciliation reports using checksum and row counts, increasing reliability and reducing ETL pipeline errors by 25% in production.
- Tracked data lineage and built metadata registries using Python scripts, enabling traceability and enhancing governance across staging and warehouse layers.
- Managed Jenkins-based CI/CD workflows to automate testing, deployment, and version control across development, staging, and production environments.
- Collaborated with QA and business analysts to perform user acceptance testing and validate transformation logic against business KPIs and domain rules.
- Participated in cloud migration POCs to evaluate Azure and GCP platforms, analyzing performance, cost, and compatibility for modernizing legacy on-premise infrastructure.

Educational Details:

Master of Science in Information Science/Studies - Trine University, Indiana, USA

Bachelor of Technology in Computer Science & Engineering - Vignan's University, Andhra Pradesh, India